NOMILA-V2 · PHASE A · STEP 0 · PROPOSAL FOR REVIEW

Step 0 Foundations

One small pull request that lays the parameter store, the canonical types and the as-of columns every later step builds on. Nothing in production reads or writes any of it. Dev behaviour is byte-identical after deploy. Built only after you say go.

base: main @ 246a98c (pulled from origin this morning)

schema flow: scripts/db-migrate.sh + drift gate (#375) next free migration number: 0043

WHAT SHIPS

1 migration · 1 schema table + 2 columns · 1 types module · 1 params helper · 1 guard · tests

About 12 files. No route, cron, prompt, engine or client file changes.

WHAT DOES NOT CHANGE

scoring_weights, computeScore, rate card, refresh, APIs

The v7.2.0 row stays the only thing the live score reads. model_params is a new table with draft rows only; drafts are never returned by the active-lookup.

WHAT YOU REVIEW HERE

Schemas, types, the parameter families, and four alpha value sets

Research TTLs and the commercial-proof source policy are proposals you asked for; they ship as draft rows or not at all, your call.

1 · Exactly what the PR changes

FILE	CHANGE
migrations/0043_intelligence_foundations.sql	New. Idempotent DDL: `model_params` table, partia index (one active row per fa nullable columns on `athlete_snapshots`. No d

scripts/db-migrate.sh	Add `0043_intelligence_found` to the hand-written list (step header range becomes 003
shared/schema.ts	`modelParams` table definiti `as_of` and `engine_versi` `athleteSnapshots`. Types exported.
server/services/intelligence/types.ts	New, types only. The §17, §2 §28.4 contracts, the eight-d enum, the deliverable taxon source-class vocabulary.
server/services/intelligence/params.ts	New. `getActiveParams(fa` `listParams(family)`, `insertDraftParams(...)` `activateParams(family, version)` (one transaction: deactivate current, activate `hashConfig`.
server/services/weightsService.ts	Guard the unconditional `config.rateEstimation.b` sort so a weights row withou dead block loads.
server/services/intelligence/params.test.ts · weightsService.test.ts	PGlite tests: one-active-per- enforced by the index; drafts to getActive; activate swaps atomically; version unique p weights row without rateEst loads.
server/services/intelligence/README.md	Two screens: what the modu vocabulary, the rule that sco imports nothing from integra pointer to the spec sections.
server/scripts/seedModelParams.ts	Optional, see section 4. Inse reviewed draft rows by hand Not run by any deploy.

> **Not in this PR, on purpose:** no writer sets `as_of` or `engine_version` (Step 1 and 2 writers do); no reader of `model_params`; no observation or feature tables (Step 1); no change to `scoring_weights` or to any cron; no environment variable; no client change. PR #309's `0034_trend_observations` is not touched and still needs its own renumber.

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2 · Schema

model_params: the versioned parameter store

Generalises the one-active-row pattern of `scoring_weights` to every parameter family the spec names: weights and sub-weights, lookup tables, priors, calibration functions, freshness TTLs, source policies, rate benchmarks. One table, one status machine, one activation rule, so every engine versions the same way (§15). `scoring_weights` is left exactly as it is; the marketability engine in Step 4 reads `model_params` and the two coexist until v7 is retired.

```
CREATE TABLE IF NOT EXISTS model_params (
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  family        text NOT NULL,          -- 'marketability' | 'signals' | 'priors' | 'cohorts' | 'rese
                                        -- | 'commercial_proof' | 'rates' | 'opportunity' | 'earnings
  version       text NOT NULL,          -- 'v1.0-beta.1', 'v0.1-alpha' — never edited in place
  status        text NOT NULL DEFAULT 'draft',
  config        jsonb NOT NULL,
  hash          text NOT NULL,          -- first 12 of sha1(canonical JSON), for eyeballing against t
  note          text,                   -- why this version exists
  created_by    text,
  created_at    timestamptz NOT NULL DEFAULT now(),
  activated_at  timestamptz,
  CONSTRAINT model_params_status_check CHECK (status IN ('draft','active','inactive','archived')),
  CONSTRAINT model_params_family_version_uq UNIQUE (family, version)
);
-- one live version per family, enforced by the database (same device as scoring_weights and prompts)
CREATE UNIQUE INDEX IF NOT EXISTS model_params_one_active_uq ON model_params (family) WHERE status =
CREATE INDEX IF NOT EXISTS model_params_family_idx ON model_params (family, created_at);
```

Why no CHECK on `family`: the same reason LOCKED_PROMPTS lives in code (0033's header). A family list a migration has to know about is one more place to keep in step with deploy DDL; the closed list lives in `types.ts` as `ParamFamily`. Why `status` and not a boolean: `draft` is the review state you asked for (never read), `inactive` is a superseded version kept for reruns (§15 "rerun historical profiles"), `archived` is retired.

athlete_snapshots: two nullable columns

```
ALTER TABLE athlete_snapshots ADD COLUMN IF NOT EXISTS as_of          timestamptz;  -- the instant th
ALTER TABLE athlete_snapshots ADD COLUMN IF NOT EXISTS engine_version text;         -- the model_para
```

Both null for every existing row and for every row the current writers insert; the score audit's history reader (`scoreHistoryRoute`) ignores unknown columns. Semantics for later steps: `as_of` defaults to `created_at` when a reader finds it null, so legacy rows need no backfill; `engine_version` null means "v7 scoring_weights active row at that time". Not added: a separate features snapshot table. That is Step 1's `social_features` plus a Step 4 feature-vector snapshot; adding an empty table now would be architecture on paper.

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3 · Types (server/services/intelligence/types.ts)

Transcribed from §17, §22.3 and §28.4, with the three places the spec is internally inconsistent resolved as noted. Types only; the file imports nothing. The rule the README states: **nothing under `server/services/intelligence/` may import from `server/lib/integrations/`** (LOCKED #9), enforced by a test that greps the import graph.

```
// Vocabulary
export type ParamFamily = 'marketability' | 'signals' | 'priors' | 'cohorts' | 'research_freshness'
                        | 'commercial_proof' | 'rates' | 'opportunity' | 'earnings';
export type SourceClass = 'connected_verified' | 'platform_public' | 'web_researched' | 'inferred' |
//  §28.4 vocabulary, plus 'inferred' from §22.3. §17's shorter set maps: verified→connected_verified
//  public→platform_public|web_researched, inferred→inferred, imputed→imputed.            (clarificat
export type MarketabilityDimension = 'athletic_authority' | 'audience_scale' | 'engagement_quality'
  | 'cultural_relevance' | 'narrative_power' | 'brand_fit' | 'commercial_proof' | 'momentum';
export type MarketabilityTier = string;   // undefined in the spec; kept open until decided (clarifi

// Evidence and signals (§5, §6, §17)
export interface EvidenceRef { evidenceId: string; sourceClass: SourceClass; observedAt: string; con
export interface NormalizedSignal {
  key: string; value: number; /* 0–100 */ confidence: number; /* 0–1 */
  source: SourceClass; evidenceIds: string[]; modelVersion: string;
  asOf: string; imputed: boolean;                       // imputed=true ⇒ value came from a prior, n
}
export interface DimensionScore { dimension: MarketabilityDimension; score: number; weight: number; c
export interface Driver { signalKey: string; dimension: MarketabilityDimension; delta: number; becaus
export interface MarketabilityResult {
  score: number; /* 0–850 */ composite: number; /* 0–100 */ potential: number | null; /* §17 field, s
  confidence: number; /* 0–100 */ tier: MarketabilityTier; dimensions: DimensionScore[]; drivers: Dri
  version: string; asOf: string;
}

// Social observations (§22.3, §28.4) — the shape Step 1's tables persist
```

```
export type MetricEntityType = 'account' | 'content' | 'audience';
export interface SocialMetricObservation {
  athleteId: string; platform: string; accountId: string; entityType: MetricEntityType; entityId: sti
  metricKey: string; value: number; unit: string; sourceClass: SourceClass; provider: string;
  observedAt: string; windowStart?: string; windowEnd?: string; confidence: number; definitionVersion
}

// Rates, opportunity, earnings (§11, §12, §17)
export type DeliverableType =
  | 'instagram_post' | 'instagram_reel' | 'instagram_story' | 'tiktok_video' | 'youtube_integration'
  | 'ugc_video' | 'product_photography' | 'brand_photoshoot' | 'short_form_package' | 'long_form_vide
  | 'appearance' | 'speaking' | 'meet_and_greet' | 'camp_clinic' | 'autograph_session'
  | 'brand_ambassador' | 'season_partnership' | 'annual_partnership'
  | 'licensing';                                    // current v3 type with no home in §11 (clari
export type RateAddOn = 'paid_media' | 'organic_usage' | 'whitelisting' | 'exclusivity' | 'extended_u
export interface RateEstimate { deliverable: DeliverableType; low: number; expected: number; high: nu
export interface OpportunityRange { conservative: number; expected: number; upside: number }
export interface EarningsPotential { conservative12m: number; expected12m: number; upside12m: number;
```

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4 · Parameter families and the alpha values you asked to review

Three draft rows are ready to insert once you have looked at the values. They can ship inside the migration (as `status = 'draft'`, inert) or through the seed script by hand. I recommend the seed script: the migration stays pure DDL, and a rejected value never becomes a deploy artefact. Activation, when it happens, follows the 0040/0042 pattern (insert inactive, swap in one transaction).

marketability · v1.0-beta.1 (frozen, transcribed from §7 and §9)

DIMENSION	WEIGHT	SUB-SIGNALS (SUB-WEIGHTS)
athletic_authority	.15	competition_level .25 · performance_results .25 · rankings_recognition .15 · national_international_status .15 · role_importance .10 · achievement_recency .10
audience_scale	.15	cross_platform_effective_reach .50 · sport_level_audience_percentile .25 · audience_growth .15 · platform_breadth .10
engagement_quality	.15	engagement_percentile .30 · reach_efficiency .25 · shares_saves_quality .15 · video_retention .10 · story_efficiency .10 · audience_authenticity .10
cultural_relevance	.10	media_volume_percentile .30 · media_source_quality .20 · search_interest .20 · social_conversation .15 · crossover_recognition .15

narrative_power	.10	distinctiveness .25 · story_depth .20 · evidence_authenticity .20 · media_resonance .15 · commercial_applicability .10 · durability .10
brand_fit	.10	category_breadth .25 · brand_safety .20 · audience_attractiveness .20 · identity_values_coherence .15 · content_versatility .10 · geographic_relevance .10
commercial_proof	.15	verified_commercial_value .30 · deal_frequency .20 · brand_quality .15 · renewals_repeat .15 · campaign_performance .10 · pricing_power .10
momentum	.10	athletic_trajectory .30 · audience_growth .25 · media_trajectory .20 · search_social_trajectory .15 · commercial_acceleration .10

mapping { kind: "power", max: 850, exponent: 0.8 } marked `alpha: true` per §9. Lookup tables, priors and cohort tables are separate families, empty until Step 4.

research_freshness · v0.1-alpha (proposal, decision 5)

TTL after which a signal family stops counting as "current" for signal computation. Evidence is never deleted; a stale item stays in history and can still feed recency-aware sub-signals (achievement_recency, durability) explicitly. Values are days.

FAMILY	TTL	WHY THIS NUMBER
identity (name, sport, teams, career status)	365	Changes on transfer, draft, retirement; athlete-asserted values override anyway.
competition_level	180	A season.
achievements	never expire; recency scored	A title is a fact forever; §7.1 already carries achievement_recency at 10%.
press_mentions (media volume, source quality)	90	Cultural relevance is about current attention; GDELT itself is a 30-day window.
brand_mentions (public partnership evidence)	180	Partnerships run seasons; older ones stay as history for repeat/renewal signals.
narrative (story, distinctiveness classifications)	180	Re-classify twice a year or on a research event.
search_interest (Trends)	30	Already a 30-day average.
social_public (discovery counts)	7	Superseded by connected data whenever present (§28.9).

social_connected (owner analytics)	2	Collected daily; two days tolerates a missed run.
commercial (validated deals, payments)	`never expire; recency scored`	Handled by the commercial_proof policy below.

commercial_proof · v0.1-alpha (proposal, decision 4)

RULE	PROPOSAL
Allowed sources for `verified_commercial_value` and `deal_frequency`	deals where `source ∈ {contract_upload, self_reported}` AND `status ∈ {accepted, completed}` ; OR any source with `payment_status = settled` (money moved through payments); OR any source with a `deal_periods` row (a contract schedule was extracted from a document).
Explicitly excluded	`source = inbox_scan` with status set by the summarizer and no settlement; `detected / negotiating / pending` anything; brand mentions from research (those stay in cultural_relevance / brand_fit, never commercial_proof: signal ownership, §21.1).
Recency window	Deals with `term_end` or `detected_at` within 24 months count fully; 24 to 48 months at half weight for `renewals_repeat` only; older excluded.
Caps (anti-circularity, §14)	At most 12 deals counted; `verified_commercial_value` saturates at the sport × level cohort p90 (the percentile, not raw dollars, is the signal) and until cohorts exist at a fixed alpha ceiling of $50,000 per 12 months; per-deal value capped at $25,000 before summing.
Where the value comes from	`deals.amount` (cash + stated product), never commission; the same rule `snapshotDealsFor` already applies.
What this means for the ten live athletes today	Mia keeps the gorjana agreement (self_reported / contract path) and the settled ones; the three `source = NULL` accepted rows and the inbox_scan accepted row would not count until settled or confirmed. Their scores do not move in Step 0; this policy is read only by the Step 4 engine.

Families created empty, filled by later steps

FAMILY	FILLED IN	HOLDS
signals	Step 4	Categorical lookup tables (competition level points, award levels, national-team status), classification-to-number maps for the bounded taxonomies (§6).
priors · cohorts	Step 1 (social) and Step 4	The prior hierarchy of §8 and the sport × level × stage × platform percentile tables.

rates · opportunity · earnings	Steps 5 and 6	Benchmarks, factor tables, add-on multipliers, volume mixes (PAV_VOLUME_MIX moves here).

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5 · How it rolls out

1. Dry run per the migration rule: `BEGIN; \i migrations/0043_intelligence_foundations.sql; ROLLBACK;` against the dev database through the proxy, then `checkSchemaDrift.ts` with the branch's schema.ts to confirm the only "missing" items are the three the migration creates.
2. PR to main: CI (typecheck, PGlite suites). Merge → `deploy-dev.yml` → `db-migrate.sh` applies 0043 → drift gate green → service replace. Watch it to green.
3. Verification after deploy: `GET /api/v1/athletes/<mia>/score` unchanged; the 04:30 refresh runs as before; `select count(*) from model_params` = 0 (or the drafts if you chose to seed).
4. Prod receives it on the next promotion through `promote.yml`; the same script runs there.

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6 · Clarifications from the spec itself (small, non-blocking)

- **C1** Two source-class vocabularies (§17 vs §22.3/§28.4). Proposed canonical: the five-value set above, with §17's four mapped onto it.
- **C2** `MarketabilityTier` is used in §17 and never defined. Left as an open string type until the display policy is decided (your decision 2).
- **C3** `MarketabilityResult.potential` (0–100) has no definition anywhere in the document. Typed as nullable; nothing will populate it until it is defined.
- **C4** The v3 rate card prices `licensing`; the §11 taxonomy has no licensing deliverable (closest are the rights add-ons). Kept in the enum so the projection adapter can carry it; needs a home in Step 5.
- **C5** §28.4 `sourceClass` lists four values while §22.3 lists five (adds `inferred`). Using five.

Your go needs to cover

- The `model_params` shape (family / version / status / config / hash, one active per family) and the two snapshot columns.

- Whether the three draft rows ship in the migration, via the seed script, or wait.
- The two alpha value sets, or edits to them. They are not read by anything until Step 4 either way.
- The source-class vocabulary (C1) and leaving tier and potential open (C2, C3).